Nonparametric SLR - Lab Activity

B.A. Hartlaub and A.S. Wagaman

Examples for Nonparametric Simple Linear Regression (includes parametric regression comparison)

Highway Runoff and Rainfall

An article in the 1998 Journal of Environmental Engineering studied highway runoff in Austin, Texas by looking at the relationship between rainfall volume (in cubic meters) and runoff volume (also in cubic meters) for different locations. The dataset is loaded next.

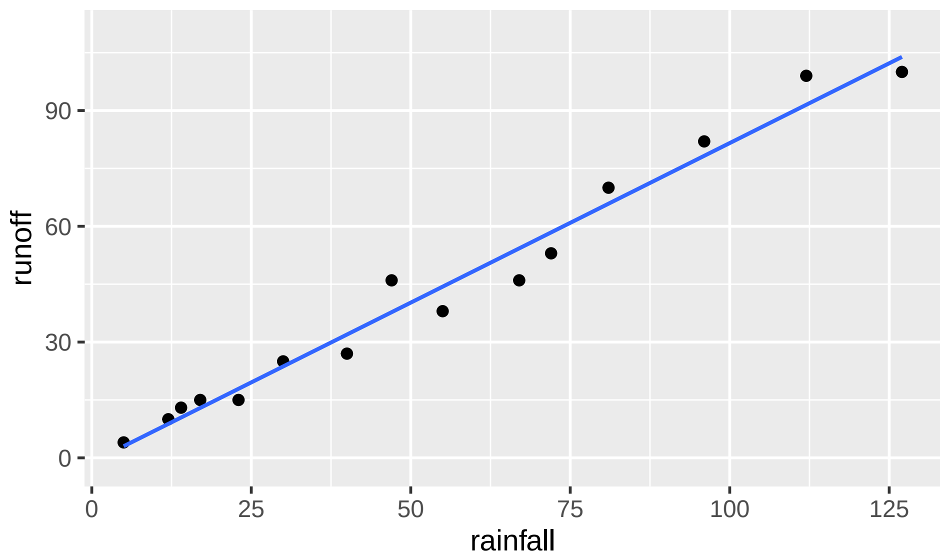
```
rainrun <-
  read.table("https://awagaman.people.amherst.edu/nonparametricstats/rainrun.txt",
             h = T)
summary(rainrun)
```

```
##     rainfall         runoff
##  Min.   :  5.0   Min.   :  4.00
##  1st Qu.: 20.0   1st Qu.: 15.00
##  Median : 47.0   Median : 38.00
##  Mean   : 53.2   Mean   : 42.87
##  3rd Qu.: 76.5   3rd Qu.: 61.50
##  Max.   :127.0   Max.   :100.00
```

```
dim(rainrun)
```

```
## [1] 15  2
```

```
ggplot(data = rainrun, aes(x = rainfall, y = runoff)) +
  geom_point() +
  geom_lm() # fits parametric regression line and shows points
```



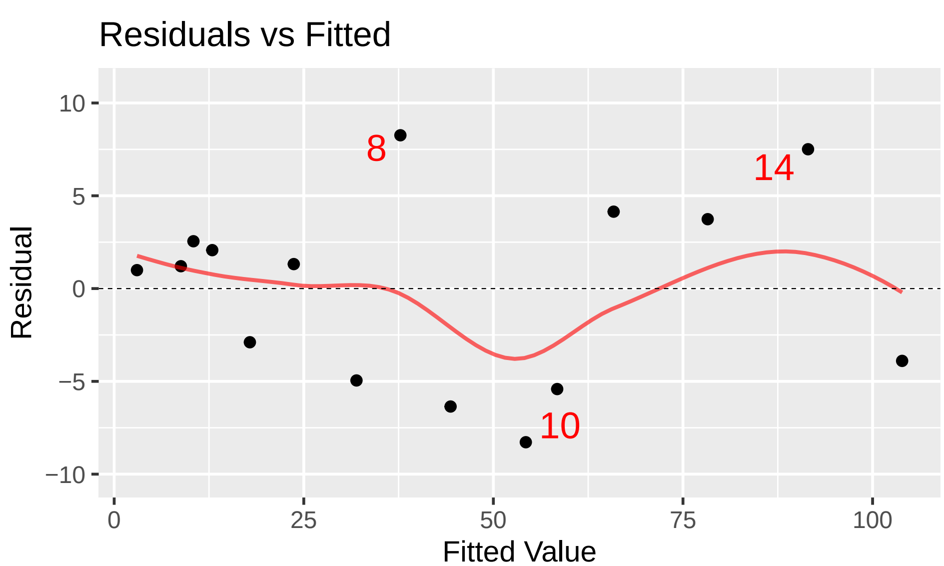
First, we'll fit the usual parametric linear regression model.

```
mod1 <- lm(runoff ~ rainfall, data = rainrun) # fit model Y ~ X
msummary(mod1) # model output
```

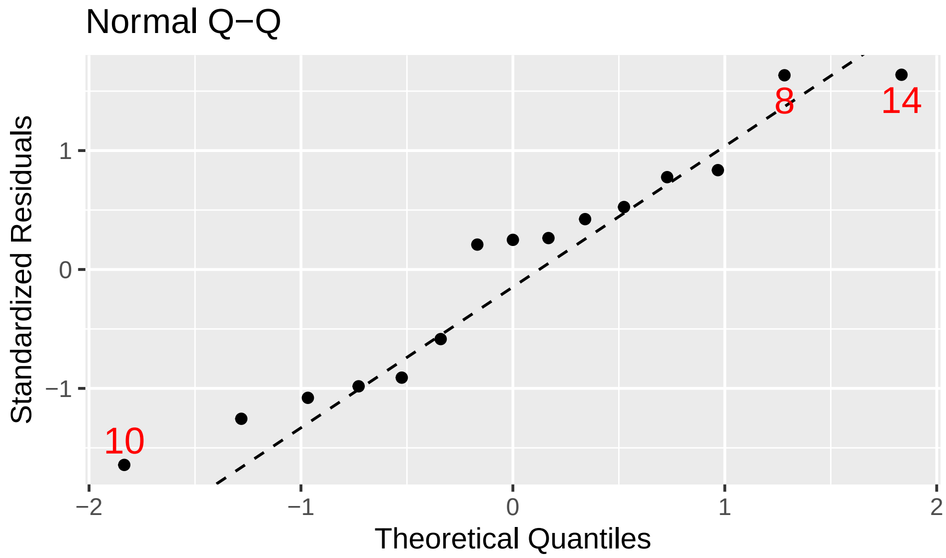
```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.12830    2.36778  -0.477    0.642
## rainfall     0.82697    0.03652  22.642  7.9e-12 ***
##
## Residual standard error: 5.24 on 13 degrees of freedom
## Multiple R-squared:  0.9753, Adjusted R-squared:  0.9734
## F-statistic: 512.7 on 1 and 13 DF,  p-value: 7.896e-12
```

```
mplot(mod1, which = 1) # residual plot to check assumption of constant variance
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```r
mplot(mod1, which = 2) # qq plot to check normality of residuals (required for errors)
```



What is the equation of the fitted parametric regression line?

SOLUTION:

Do you see any major issues with the assumptions for the parametric test here?

SOLUTION:

Is there a significant linear relationship between rainfall and runoff volume?

SOLUTION:

Now, compare the results to what you find using a nonparametric linear regression model. You will need to provide the code to fit the model.

SOLUTION:

```r
# use class examples to help!
```

What is the equation of the fitted nonparametric regression line?

SOLUTION:

Is there a significant linear relationship between rainfall and runoff volume based on the nonparametric procedure? Is this result consistent with the parametric test results?

SOLUTION:

Predict the runoff (response) when the rainfall value is 42, using both models. How should you interpret the predicted values? (There should be a difference.)

SOLUTION:

Lichen

An article in Environmental Monitoring and Assessment in 1993 explored the relationship between NO3 wet desposition (grams per cubic meter) and lichen (percent dry weight) in order to try to predict lichen dryness using NO3. The dataset is loaded next.

```r
lichen <-
  read.table("https://awagaman.people.amherst.edu/nonparametricstats/lichen.txt",
             h = T)
summary(lichen)
```

```
##       no3             lichen
##  Min.   :0.0500   Min.   :0.4800
##  1st Qu.:0.1200   1st Qu.:0.5200
##  Median :0.4200   Median :0.8600
##  Mean   :0.4554   Mean   :0.8054
##  3rd Qu.:0.6800   3rd Qu.:1.0000
##  Max.   :0.9200   Max.   :1.7000
```

```r
dim(lichen)
```

```
## [1] 13  2
```

If the goal is to predict lichen dryness using NO3, which variable should be the response and which the predictor?

SOLUTION:

Fit a parametric regression model. Then address the questions below.

SOLUTION:

What is the equation of the fitted parametric regression line?

SOLUTION:

Do you see any major issues with the assumptions for the parametric test here?

SOLUTION:

Is there a significant linear relationship between the two variables?

SOLUTION:

Now, compare the results to what you find using a nonparametric linear regression model. Fit the model first.

SOLUTION:

What is the equation of the fitted nonparametric regression line?

SOLUTION:

Is there a significant linear relationship between the two variables based on the nonparametric procedure? Is this result consistent with the parametric test results?

SOLUTION:

Would it be appropriate to predict the NO3 value when the lichen percentage is 50 percent using the regression lines obtained above? Explain.

SOLUTION:

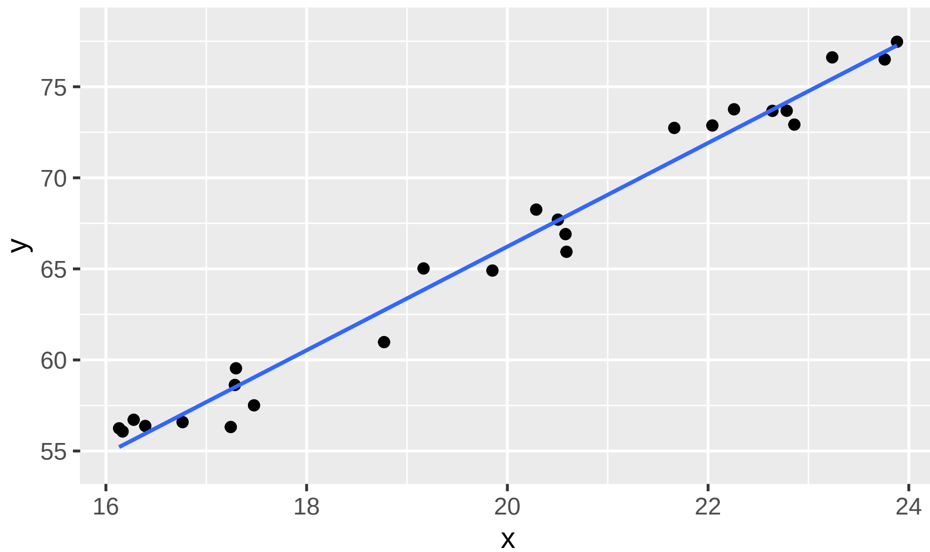
Comparing Parametric and Nonparametric SLR

We want to investigate settings where the differences between the nonparametric and parametric procedures can be observed. This lies in studying the distribution of errors, which we can explore via simulation.

Let's consider a model where $Y = 3X + 6 + error$, and then tinker with what distribution we use for the error. Remember that the errors are supposed to have median 0, but we can change the spread and shape of the distribution quite a bit and still satisfy this. Most of our examples below also involve symmetric distributions, but that doesn't have to be the case.

Example 1 - Uniform Errors

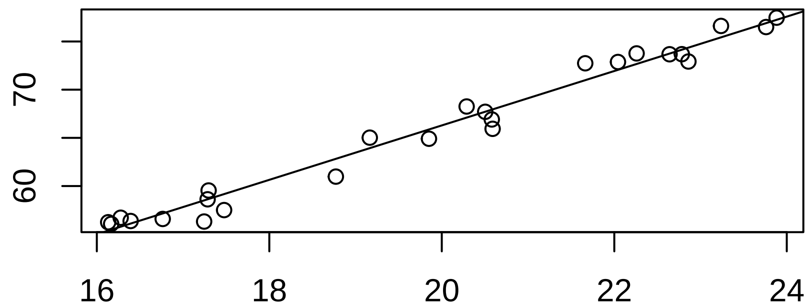
```
set.seed(52) # you can change this!
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + runif(25, -2, 2) # Y=3X+6+ error; error is uniform(-2,2)
mydata <- data.frame(x, y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.31150    1.77586   5.243 2.56e-05 ***
## x            2.84558    0.08875  32.063  < 2e-16 ***
##
## Residual standard error: 1.167 on 23 degrees of freedom
## Multiple R-squared:  0.9781, Adjusted R-squared:  0.9772
## F-statistic:  1028 on 1 and 23 DF,  p-value: < 2.2e-16
```

```
with(mydata, theil(x, y, beta.0 = 0, type = "t"))
```



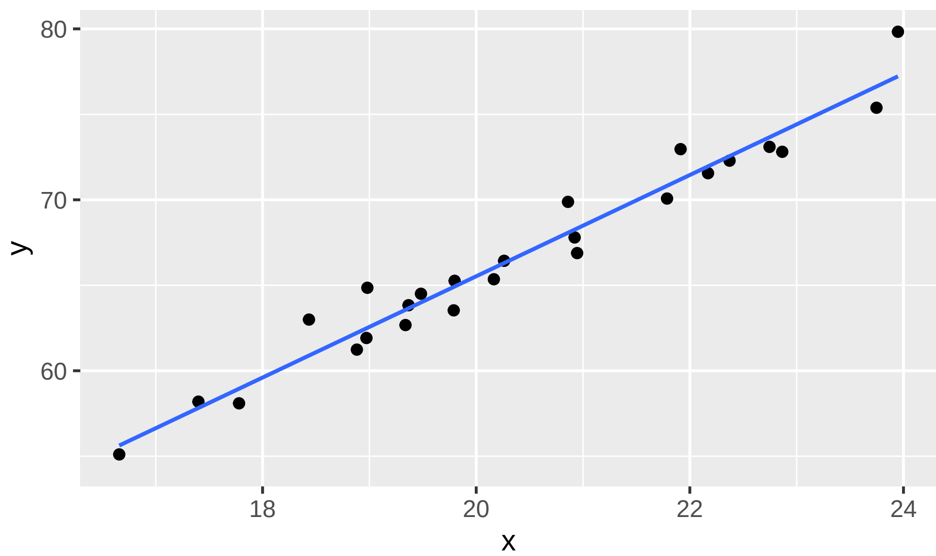

```
## Alternative: beta not equal to 0
## C = 258, C.bar = 0.86, P = 0
## beta.hat = 2.823
## alpha.hat = 9.838
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 2.623, 2.994
```

How do the parametric and nonparametric lines compare here? (Look at both plots and their coefficients!) Is this what you expected?

SOLUTION:

In the code below, increase the spread of the errors by using a wider range for the uniform distribution (like -5 to 5, etc.)

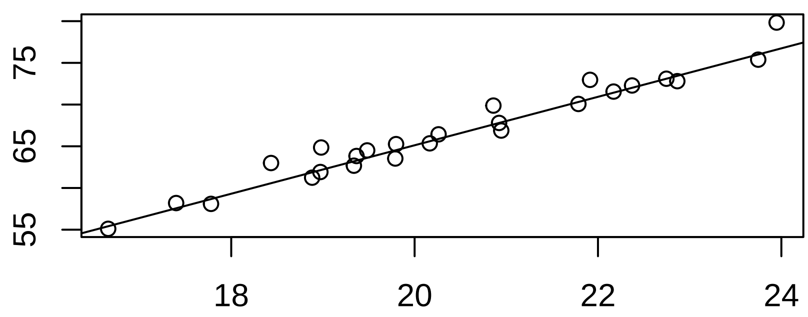
```
set.seed(22) # you can change this!
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + runif(25, -2, 2) # change -2 and 2 to wider range, centered at 0
mydata <- data.frame(x, y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```r
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.279      2.661    2.36   0.0271 *
## x              2.962      0.130   22.79   <2e-16 ***
##
## Residual standard error: 1.242 on 23 degrees of freedom
## Multiple R-squared:  0.9576, Adjusted R-squared:  0.9558
## F-statistic: 519.5 on 1 and 23 DF,  p-value: < 2.2e-16
```

```r
with(mydata, theil(x, y, beta.0 = 0, type = "t"))
```




```
## Alternative: beta not equal to 0
## C = 266, C.bar = 0.887, P = 0
## beta.hat = 2.905
## alpha.hat = 7.018
```

```
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 2.685, 3.144
```

How do the parametric and nonparametric lines compare here? Is this what you expected?

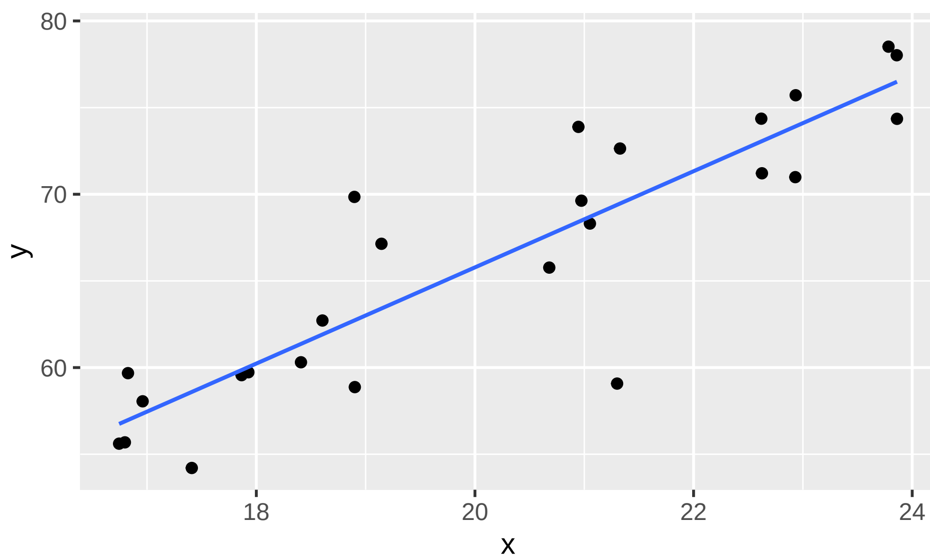
SOLUTION:

Note that this is only a single data set comparison (under each distribution of errors). To do a more appropriate comparison, you'd need to do multiple runs here, comparing multiple slope coefficients across both models (or figure out some other measure you want to use for assessment).

Example 2 - Double Exponential Errors

We've seen before that the parametric and nonparametric procedures are often near equivalent in settings with either underlying normal distributions or uniform distributions, so let's get a little bit more extreme.

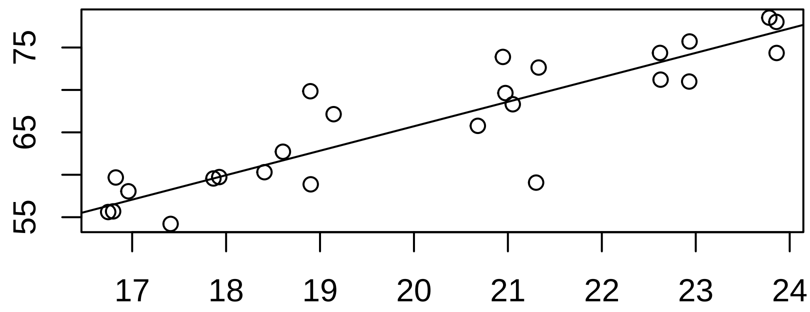
```r
set.seed(29) # you can change this!
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + rlaplace(25, 0, 2) # Need center at 0, can change scale parameter
mydata <- data.frame(x, y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```r
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.2940     6.0012   1.715   0.0997 .
## x             2.7744     0.2959   9.375 2.55e-09 ***
##
## Residual standard error: 3.562 on 23 degrees of freedom
## Multiple R-squared:  0.7926, Adjusted R-squared:  0.7836
## F-statistic: 87.89 on 1 and 23 DF,  p-value: 2.548e-09
```

```r
with(mydata, theil(x, y, beta.0 = 0, type = "t")) # nonparametric plot is part of output
```



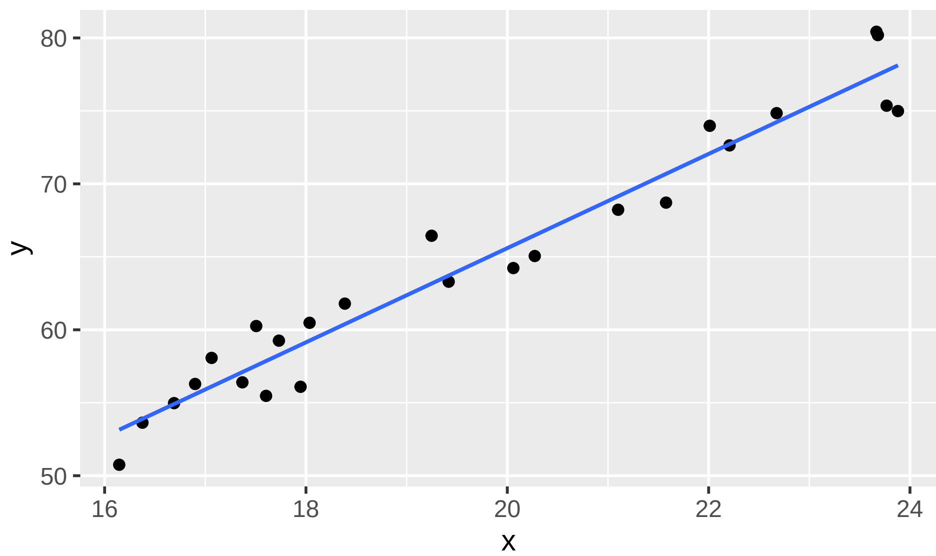

```
## Alternative: beta not equal to 0
## C = 212, C.bar = 0.707, P = 0
## beta.hat = 2.883
## alpha.hat = 8.065
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 2.286, 3.3
```

How do the parametric and nonparametric lines compare here? Is this what you expected?

SOLUTION:

Once again, up the spread of the errors to see what happens. In the code below, increase the spread of the errors by using a larger value for the scale parameter

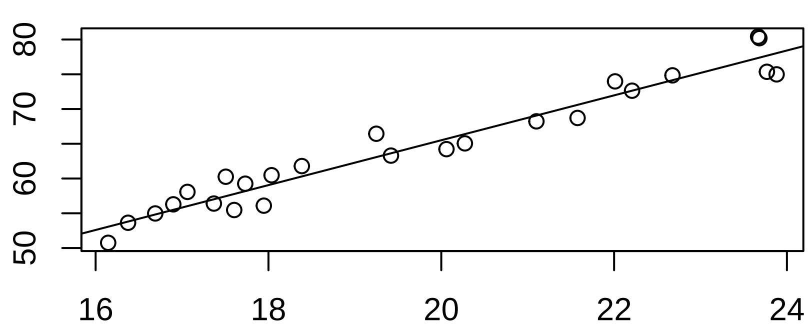
```
set.seed(51) # you can change this!
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + rlaplace(25, 0, 2) # Change scale parameter to be > 2
mydata <- data.frame(x, y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```r
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.0350     3.0841   0.336     0.74
## x             3.2280     0.1556  20.745   <2e-16 ***
##
## Residual standard error: 2.004 on 23 degrees of freedom
## Multiple R-squared:  0.9493, Adjusted R-squared:  0.9471
## F-statistic: 430.4 on 1 and 23 DF,  p-value: < 2.2e-16
```

```r
with(mydata, theil(x, y, beta.0 = 0, type = "t")) # nonparametric plot is part of output
```




```
## Alternative: beta not equal to 0
## C = 258, C.bar = 0.86, P = 0
## beta.hat = 3.228
## alpha.hat = 0.95
```

```
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 2.83, 3.647
```

How do the parametric and nonparametric lines compare here? Is this what you expected?

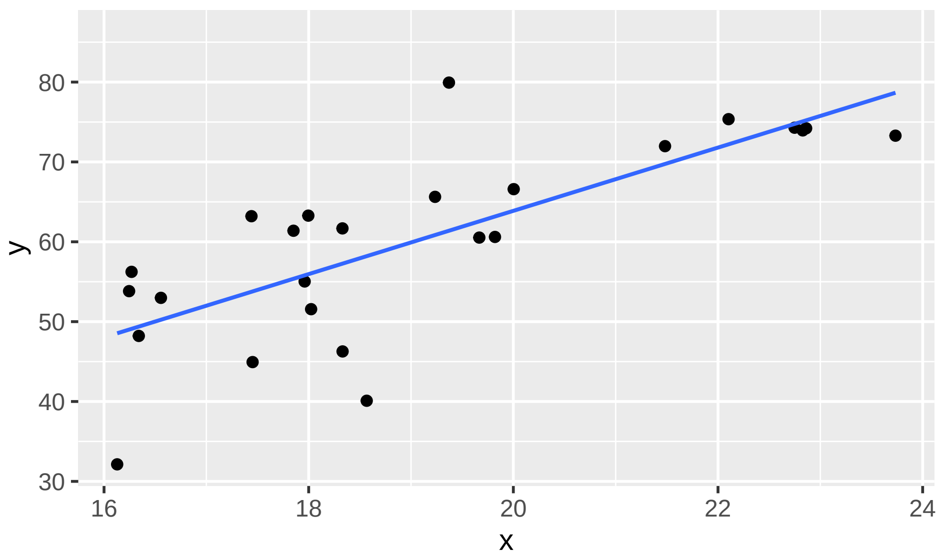
SOLUTION:

Note that if the error variance starts getting too large, both lines will start to have issues - the relationship is weaker.

Example 3 - Cauchy Errors

One last example before we look at the effect of outliers, though this example helps to illustrate that too. What happens if we use a Cauchy distribution for the errors?

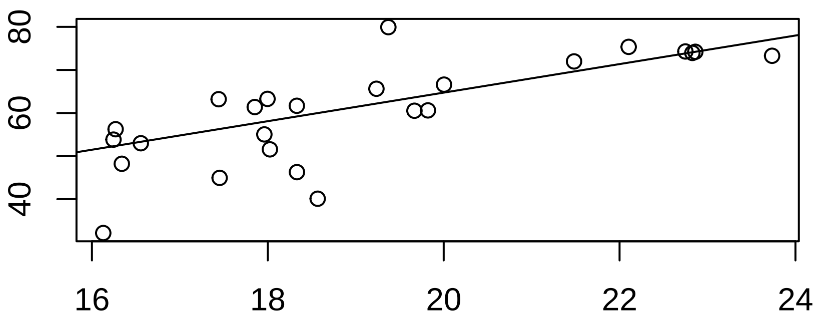
```r
set.seed(87) # you can change this!
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + rcauchy(25, 0, 2) # Need center at 0, can change scale parameter
mydata <- data.frame(x ,y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```r
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -15.332     13.730  -1.117    0.276
## x              3.960      0.714   5.547 1.21e-05 ***
##
## Residual standard error: 8.157 on 23 degrees of freedom
## Multiple R-squared:  0.5722, Adjusted R-squared:  0.5536
## F-statistic: 30.77 on 1 and 23 DF,  p-value: 1.214e-05
```

```r
with(mydata, theil(x, y, beta.0 = 0, type = "t")) # nonparametric plot is part of output
```




```
## Alternative: beta not equal to 0
## C = 156, C.bar = 0.52, P = 0
## beta.hat = 3.314
## alpha.hat = -1.558
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 2.335, 4.812
```

How do the parametric and nonparametric lines compare here? Is this what you expected?

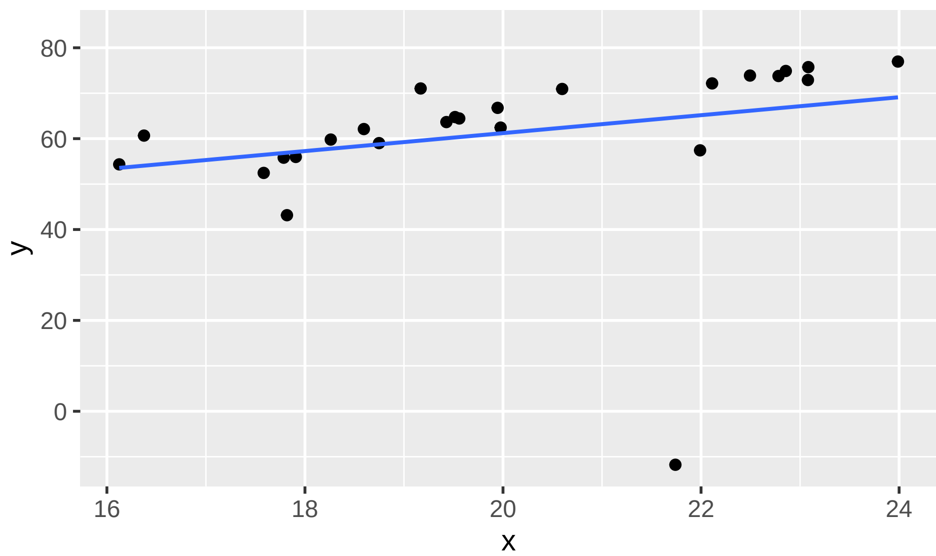
SOLUTION:

Are there outliers here? What is affecting the performance of the parametric regression line?

SOLUTION:

One last time, up the spread of the errors to see what happens. In the code below, increase the spread of the errors by using a larger value for the scale parameter. Again, if it goes too large, the relationship can become weak enough that both methods will start to have trouble.

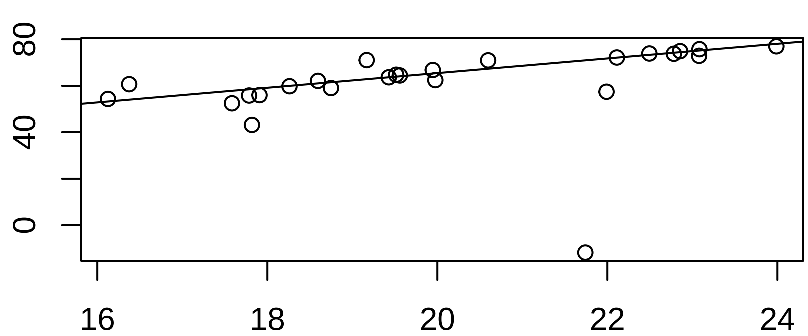
```
set.seed(34) # you can change this!
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + rcauchy(25, 0, 2) # change scale parameter > 2
mydata <- data.frame(x ,y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   21.719     31.162   0.697    0.493
## x              1.975      1.544   1.279    0.214
##
## Residual standard error: 17.25 on 23 degrees of freedom
## Multiple R-squared:  0.06641,    Adjusted R-squared:  0.02582
## F-statistic: 1.636 on 1 and 23 DF,  p-value: 0.2136
```

```
with(mydata, theil(x, y, beta.0 = 0, type = "t")) # nonparametric plot is part of output
```




```
## Alternative: beta not equal to 0
## C = 192, C.bar = 0.64, P = 0
## beta.hat = 3.152
## alpha.hat = 2.412
```

```
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 2.534, 3.839
```

How do the parametric and nonparametric lines compare here? Is this what you expected?

SOLUTION:

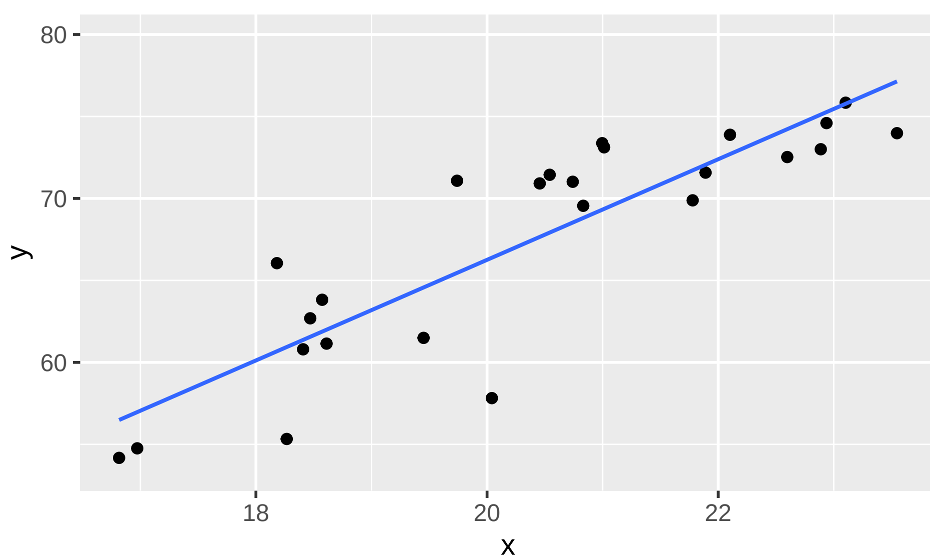
Are there outliers here? What is affecting the performance of the parametric regression line?

SOLUTION:

Example 4 - Effect of Outliers

The Cauchy examples should have started to suggest that the parametric procedures are much more sensitive to outliers than the nonparametric procedures. Here, we investigate that more directly. We want to start with a relationship where both procedures do fairly well. We can even use a normal distribution for the errors.

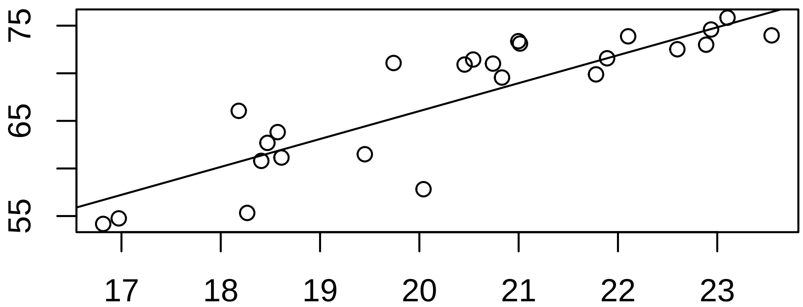
```r
set.seed(76) # keep this, or remember to change both here and below
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + rnorm(25, 0, 3) # Y=3X+6+ error; error is N(0,3)
mydata <- data.frame(x, y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```r
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.904      7.341   0.668    0.511
## x              3.068      0.359   8.544 1.36e-08 ***
##
## Residual standard error: 3.452 on 23 degrees of freedom
## Multiple R-squared:  0.7604, Adjusted R-squared:   0.75
## F-statistic: 73.01 on 1 and 23 DF,  p-value: 1.361e-08
```

```r
with(mydata, theil(x, y, beta.0 = 0, type = "t")) # nonparametric plot is part of output
```




```
## Alternative: beta not equal to 0
## C = 218, C.bar = 0.727, P = 0
## beta.hat = 2.933
## alpha.hat = 7.375
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 1.911, 3.824
```

How do the parametric and nonparametric lines compare here? Is this what you expected?

SOLUTION:

P.S. It should be good that the nonparametric line is not doing badly in a setting where the parametric line is expected to do well.

Now, let's create an influential point by modifying an existing data point. This is a point that we think will affect the regression line strongly because it is off the general pattern.

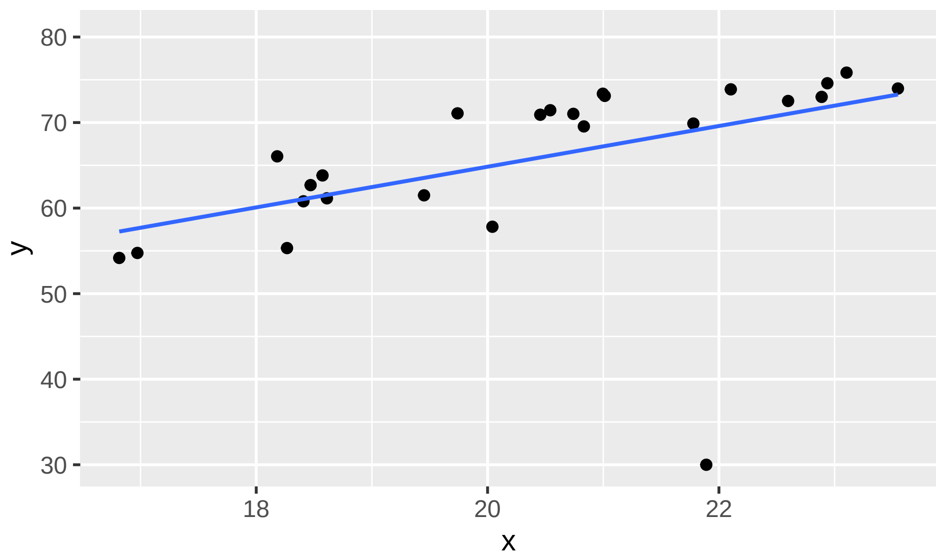
The first and second observations were:

```r
head(mydata, 2)
```

```
##          x        y
## 1 20.99668 73.37156
## 2 21.89073 71.57486
```

The second observation is very close to both lines, so let's tinker with that one. We will leave it at that X value, but change the y value. Instead of 71.57, let's drop the Y value to 30, and see what happens.

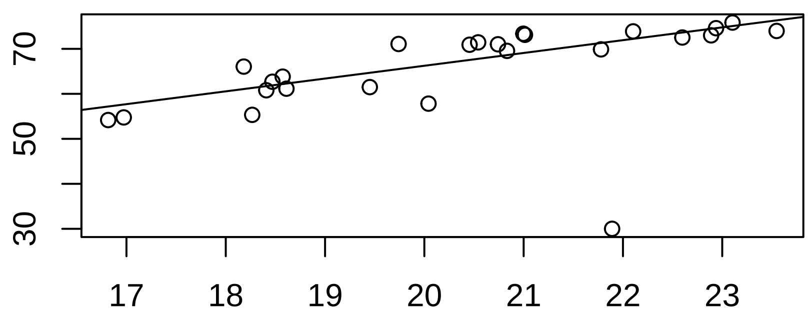
```r
# brute force way to change
mydata[2, 2] <- 30
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```r
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  17.2657    19.4769   0.886   0.3845
## x             2.3787     0.9525   2.497   0.0201 *
##
## Residual standard error: 9.159 on 23 degrees of freedom
## Multiple R-squared:  0.2133, Adjusted R-squared:  0.1791
## F-statistic: 6.237 on 1 and 23 DF,  p-value: 0.02011
```

```r
with(mydata, theil(x, y, beta.0 = 0, type = "t")) # nonparametric plot is part of output
```




```
## Alternative: beta not equal to 0
## C = 186, C.bar = 0.62, P = 0
## beta.hat = 2.845
## alpha.hat = 9.347
```

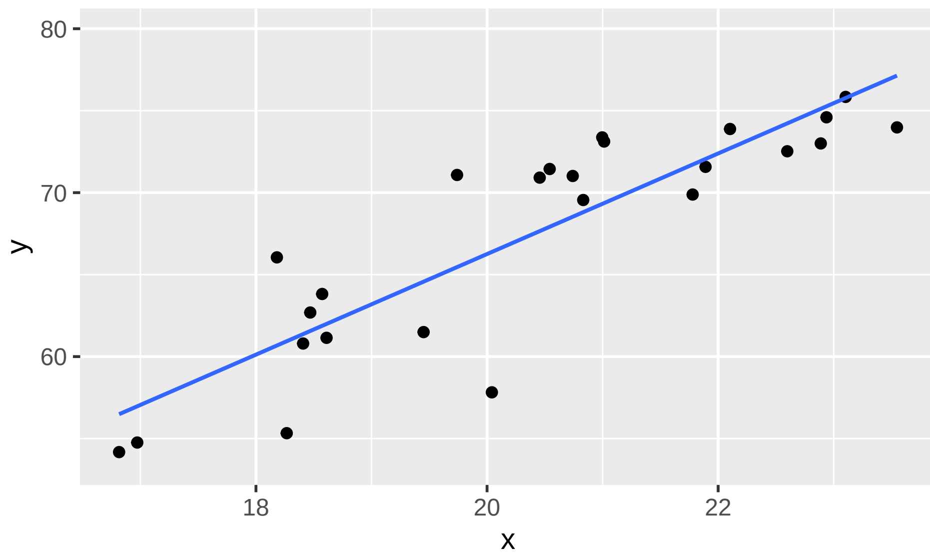
```
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 1.683, 3.828
```

What effect did you observe in the regression lines? Which line was more affected by the unusual point?

SOLUTION:

Points further from the mean of the x's have more leverage to influence the regression line. Let's reload the data and tinker with one of the furthest points out.

```
set.seed(76) # match what you started Cauchy section with
x <- runif(25, 16, 24) # 25 obs from a Unif(16,24)
y <- 3*x + 6 + rnorm(25, 0, 3) # Y=3X+6+ error; error is N(0,3)
mydata <- data.frame(x, y)
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```
max(x)
```

```
## [1] 23.54757
```

```
which(x>23.545)
```

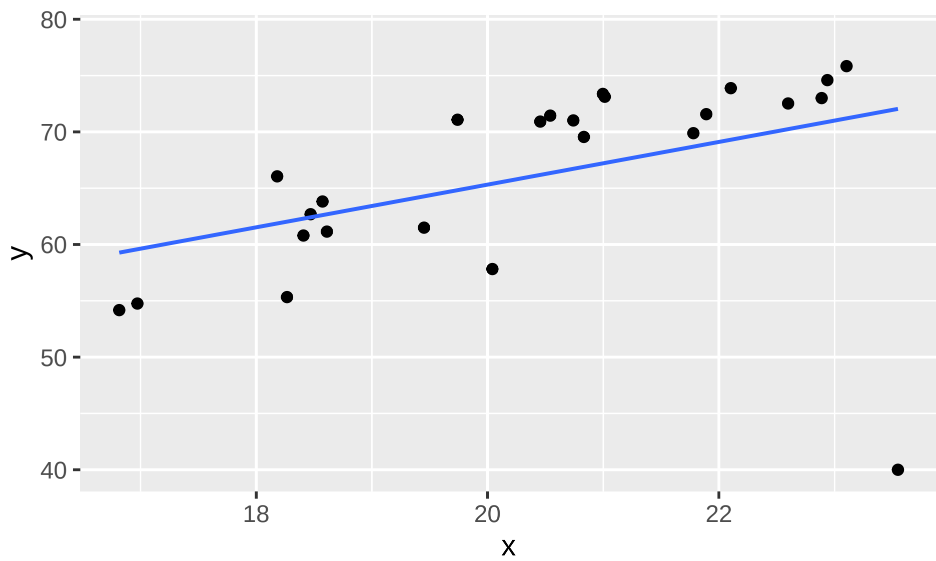
```
## [1] 13
```

```
mydata[13, ]
```

```
##           x        y
## 13 23.54757 73.98012
```

The maximum x value (we could also do minimum) is observation 13. Let's change (drop) it's Y value to 40 and see what happens.

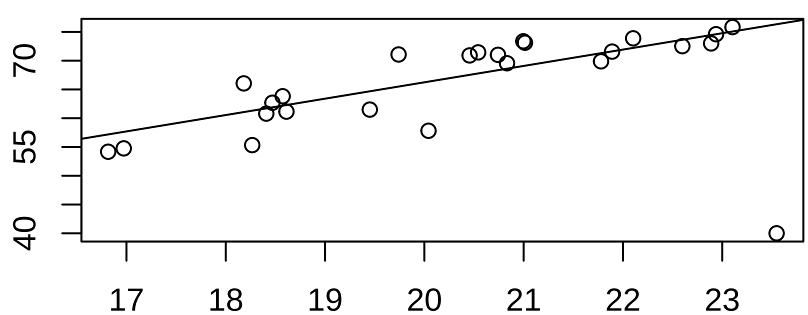
```
mydata[13, 2] <- 40
ggplot(mydata, aes(x = x, y = y)) + geom_point() + geom_lm() # parametric fit
```




```r
fm <- lm(y ~ x, data = mydata)
msummary(fm)
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  27.4015    17.0013   1.612   0.1207
## x             1.8957     0.8314   2.280   0.0322 *
##
## Residual standard error: 7.995 on 23 degrees of freedom
## Multiple R-squared:  0.1844, Adjusted R-squared:  0.1489
## F-statistic: 5.199 on 1 and 23 DF,  p-value: 0.03219
```

```r
with(mydata, theil(x, y, beta.0 = 0, type = "t")) # nonparametric plot is part of output
```




```
## Alternative: beta not equal to 0
## C = 174, C.bar = 0.58, P = 0
## beta.hat = 2.845
## alpha.hat = 9.347
```

```
##
## 1 - alpha = 0.95 two-sided CI for beta:
## 1.644, 3.811
```

What effect did you observe in the regression lines? Which line was more affected by the unusual point?

SOLUTION:

What if we dropped the y-value even further there to something like 20? Which line would you expect to be more affected?

SOLUTION:

Feel free to check to verify your thoughts.

You should be able to describe the benefits of nonparametric procedures based on these examples, as well as understand a bit more about how the parametric procedures can be affected by unusual points or non-normal errors.

Data Sources

Barrett, Michael E., et al. "Characterization of highway runoff in Austin, Texas, area." Journal of Environmental Engineering 124.2 (1998): 131-137.

Bruteig, Inga E. "The epiphytic lichen Hypogymnia physodes as a biomonitor of atmospheric nitrogen and sulphur deposition in Norway." Environmental monitoring and assessment 26.1 (1993): 27-47.

Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to Nonparametric Statistical Methods (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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